

The expected number of faces of a non-orientable random embedding of K_n is at most $\ln n + O(1)$

A proof of Conjecture 9.5 of Campion Loth, Halasz, Masařík, Mohar and Šámal

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Machine-generated note

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Abstract

A non-orientable random embedding of a graph, in the sense of Campion Loth, Halasz, Masařík, Mohar and Šámal (arXiv:2211.01032), chooses a uniformly random rotation system together with independent fair signs on the edges; F_n denotes the resulting number of faces of K_n . They conjectured (Conjecture 9.5) that $\mathbb{E}F_n \leq \ln n + O(1)$. We prove this, in the explicit form $\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O((\ln n)^2/n)$ with γ Euler’s constant, together with a finite- n inequality valid for all $n \geq 21$. The proof splits the faces according to whether they pass through a fixed vertex o : the faces through o are counted by exposing the single local matching at o pair by pair, and the faces avoiding o are controlled by a trace process in which the untouched edge towards o supplies an escape probability at every step, which yields an exponential survival bound and a geometric bound on repeated visits to any vertex. Conditioning on the surface being non-orientable does not change the conclusion.

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1 Introduction

Random embeddings. A *rotation system* of a finite graph G assigns to each vertex a cyclic order of its incident edge-ends; it determines a cellular embedding of G in an orientable surface, and the *random embedding* of G chooses the cyclic orders independently and uniformly. The companion note on Conjecture 9.3 of the same paper [2] sets out these conventions in detail, and we refer to it for the orientable model. For non-orientable surfaces one uses an *embedding scheme* [3, 4]: a rotation system together with a *signature* $\varepsilon: E(G) \rightarrow \{+1, -1\}$, where $\varepsilon(e) = -1$ means that the band attached along e is twisted. Every embedding scheme determines a cellular embedding in a closed surface, whose faces are found by the usual face-tracing procedure, and the surface is orientable if and only if the signature is switching-equivalent to the all-positive one.

Campion Loth, Halasz, Masařík, Mohar and Šámal [1] proved, for the orientable random embedding of the complete graph, that $\frac{1}{2} \ln n - 2 < \mathbb{E}F(K_n) \leq 3.65 \ln n + o(1)$, and in their closing section (“Open Problems”) they turned to non-orientable surfaces. The referee checked the following two quotations against the arXiv v3 T_EX source ([Appendix B](#)). The model is introduced as follows:

One natural way to define a random embedding of a graph on a non-orientable surface is to randomly choose a rotation system, and randomly choose a signature for all the edges in the graph, with probability $1/2$ of being either sign. From data, we expect a similar result to hold for non-orientable random embeddings of K_n under this definition.

The conjecture then reads:

Conjecture 1.1 ([1, Conjecture 9.5]). The expected number of faces in a non-orientable random embedding of the complete graph K_n is at most $\ln(n) + O(1)$.

The catalog page from which the problem was posed records that the conjecture first appeared in the April 2025 version of the paper, that it is supported by the authors' simulations, and that no subsequent work on it was found; the referee's own literature search (Appendix B) confirmed this in September 2026.

Conventions. Throughout, $n \geq 3$, $d = n - 1$ is the degree of K_n , and all logarithms are natural; $H_m = \sum_{i \leq m} 1/i$ is the harmonic number and γ is Euler's constant. The *non-orientable random embedding* of K_n is the embedding scheme (π, ε) in which the cyclic orders π_u are independent and uniform among the $(d - 1)!$ cyclic orders at each vertex u , and the signs $\varepsilon(e)$ are independent, fair, and independent of π ; nothing is conditioned. This is exactly the model of the quotation above. F_n denotes the number of faces of the resulting embedding. Since the model also produces orientable embeddings with a very small probability, Proposition 5.1 shows that conditioning on non-orientability changes nothing. We use that K_n is a *simple* graph: there is exactly one edge between any two vertices. This hypothesis is load-bearing (Remark 6.2).

Theorem 1.2 (Main theorem). Let $R = \lceil 4 \log_2 n \rceil$, $D = n - 2 - R$ and $q = 1 - \frac{1}{n-2}$. For every $n \geq 21$ (so that $D \geq 1$),

$$\mathbb{E}F_n \leq 1 + \frac{1}{2}H_{n-2} + \frac{(n-1)(n-2)}{2D^2} q^{-2} \left(\ln(n-2) - q - \frac{q^2}{2} \right) + \frac{1}{n}. \quad (1)$$

Consequently, as $n \rightarrow \infty$,

$$\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right). \quad (2)$$

Corollary 1.3. Conjecture 1.1 holds: $\mathbb{E}F_n \leq \ln n + O(1)$. The same bound holds for the expectation conditioned on the embedding being non-orientable (Proposition 5.1).

Proof. (2) gives the bound for $n \geq n_1$ for some n_1 , and $F_n \leq n(n-1)/3$ deterministically for $n \geq 3$ (every face has length at least 3 by Lemma 2.3, and the face lengths sum to $n(n-1)$), so the finitely many remaining values are absorbed into the $O(1)$. The conditional statement is Proposition 5.1. $\square$

Idea of proof. The embedding is encoded by a flag model (Section 2): every incidence carries two flags, an edge matching A joins corresponding flags at the two ends of every edge, and at each vertex u a local matching V_u describes the arcs of the vertex disc between consecutive attachments; faces are the components of $A \cup V$. Absorbing the signs into per-incidence relabellings makes A canonical and the V_u independent and uniform on the set $\mathcal{H}_d$ of matchings that close up with the incidence pairs into a single $2d$ -cycle. The key tool is an exposure lemma: after $r \leq d - 2$ pairs of V_u have been revealed, the mate of any unmatched flag is uniform over exactly $2(d - r - 1)$ candidates.

Fix a vertex o and split $F_n = F_n^{\text{hit}} + F_n^{\text{avoid}}$ into faces meeting and avoiding o . Revealing all V_u with $u \neq o$ leaves the avoiding faces as closed cycles and pairs up the $2d$ flags at o by paths; exposing V_o pair by pair, each step closes a face with probability at most $1/(2(d - r - 1))$, whence $\mathbb{E}F_n^{\text{hit}} \leq 1 + \frac{1}{2}H_{n-2}$ (Lemma 3.1). For the avoiding faces we follow one face from a starting flag and stop on reaching o . Because the two flags of the edge towards o are untouched while the trace avoids o , each step moves to o with probability exactly $1/(n - 2 - r_u)$, which is at least the probability of moving to any other prescribed vertex (Lemma 4.1). This gives an exponential survival bound $\Pr(T > t) \leq q^t$ and, via a supermartingale, $\Pr(\geq L \text{ visits to } w) \leq 2^{1-L}$ for every vertex w (Lemma 4.2). Faces loading some vertex more than $R = \lceil 4 \log_2 n \rceil$ times therefore contribute at most $1/n$ in expectation (Lemma 4.3), and for the remaining *good* faces the last two steps of the trace have probability at most $1/D$ and $1/(2D)$ respectively, giving $\Pr(\text{good avoiding face of length } k) \leq q^{k-2}/(2D^2)$ uniformly in k (Lemma 4.4). Summing over lengths with the rooting identity $F = \sum_a 1/(2L(a))$ yields $\mathbb{E}F_n^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4} + o(1)$ (Proposition 4.5). [full proof]

2 The flag model

Let $n \geq 3$ and $d = n - 1$. For every ordered pair of distinct vertices (u, v) and every $s \in \{0, 1\}$ there is a *flag* (u, v, s) , at u , on the edge uv ; there are $2d$ flags at each vertex and $2n(n - 1)$ in total. Three perfect matchings on the flags are used:

- J_u pairs $(u, v, 0)$ with $(u, v, 1)$ for every $v \neq u$ (the two flags of one incidence); $J = \bigcup_u J_u$.
- A pairs (u, v, s) with (v, u, s) (the two sides of the band of the edge uv).
- V_u is a perfect matching on the flags at u such that $J_u \cup V_u$ is a single $2d$ -cycle; $V = \bigcup_u V_u$.
Let $\mathcal{H}_d$ be the set of such matchings.

Every flag has degree one in A and one in V , so $A \cup V$ is a disjoint union of cycles alternating between A - and V -edges. A cycle with k edges of each kind has $2k$ flags and is said to have *length* k ; the lengths of all cycles sum to $|A| = n(n - 1)$.

Lemma 2.1 (Flag model). *Let (π, ε) be the non-orientable random embedding of K_n . There is a coupling with matchings $(V_u)_{u \in V(K_n)}$, independent and uniform on $\mathcal{H}_d$, such that the faces of (π, ε) correspond bijectively, and length-preservingly, to the cycles of $A \cup V$. Moreover $|\mathcal{H}_d| = 2^{d-1}(d - 1)!$.*

Idea of proof. In the ribbon graph of an embedding scheme the two flags of an incidence are the two corners at which the band meets the vertex disc, the arcs of the disc boundary between consecutive attachments give a local matching in $\mathcal{H}_d$, and the band of $e = uv$ joins (u, v, s) to $(v, u, s \oplus \tau_e)$ where the twist bit τ_e is $\varepsilon(e)$ up to a fixed convention. Write the fair twist bits as $\tau_e = b_{u,e} \oplus b_{v,e}$ with independent fair bits $b_{u,e}$, one per *incidence*, and relabel the two flags of each incidence (u, e) according to $b_{u,e}$. The band matching becomes the canonical A , and the local matching at u becomes the image of $(\pi_u, (b_{u,e})_e)$ under a two-to-one map onto $\mathcal{H}_d$ (the two directions around the cycle), so it is uniform on $\mathcal{H}_d$ and $|\mathcal{H}_d| = 2^d(d - 1)!/2$. [\[full proof\]](#)

Lemma 2.2 (Exposure lemma). *Let V_u be uniform on $\mathcal{H}_d$ and let $Q \subseteq V_u$ consist of r pairs. Then $J_u \cup Q$ is a disjoint union of $d - r$ paths whose endpoints are exactly the flags at u not matched by Q . Conditionally on $\{Q \subseteq V_u\}$, for any unmatched flag x :*

- (i) *if $r \leq d - 2$, the V_u -mate of x is uniform over the $2(d - r - 1)$ unmatched flags other than x and the other endpoint of the path of $J_u \cup Q$ containing x ;*
- (ii) *if $r = d - 1$, the mate of x is that other endpoint.*

The same holds when Q is built adaptively, each pair being revealed by querying the mate of a flag chosen as a function of the pairs revealed so far and of randomness independent of V_u .

Idea of proof. Contracting each path of $J_u \cup Q$ to a single J -edge is a bijection between $\{W \in \mathcal{H}_d : Q \subseteq W\}$ and $\mathcal{H}_{d-r}$. Pairing x with the other endpoint of its own path closes a proper cycle and is forbidden; pairing it with any other unmatched flag merges two paths and leaves exactly $|\mathcal{H}_{d-r-1}|$ completions, the same number for every choice. [\[full proof\]](#)

Lemma 2.3 (No short faces). *For $n \geq 3$ no face has length 1 or 2.*

Proof. A length-one cycle $x A y V x$ would have the pair $\{x, y\}$ in both A and V ; but an A -pair joins flags at two different vertices and a V -pair joins flags at the same vertex. (Equivalently, K_n has no loops.) A length-two cycle $x_1 A x_2 V x_3 A x_4 V x_1$ uses two A -edges; since x_2 and x_3 are at the same vertex, the two A -edges join the same pair of vertices, hence (as K_n is simple) are the two sides of one edge uv , so $\{x_2, x_3\} = \{(v, u, 0), (v, u, 1)\}$ and the V -pair $x_2 x_3$ is a J_v -edge. Then $J_v \cup V_v$ contains the 2-cycle formed by this edge, contradicting $V_v \in \mathcal{H}_d$ for $d \geq 2$. $\square$

3 Faces through a fixed vertex

Fix a vertex o and write $F_n = F_n^{\text{hit}} + F_n^{\text{avoid}}$, where F_n^{hit} counts the faces containing a flag at o and F_n^{avoid} the others. Note that a face avoiding o contains no flag on any edge incident with o (such a flag is A -matched to a flag at o). Let S be the set of flags on edges not incident with o ; then

$$|S| = 4 \binom{n-1}{2} = 2(n-1)(n-2). \quad (3)$$

Lemma 3.1 (Faces through o). $\mathbb{E}F_n^{\text{hit}} \leq 1 + \frac{1}{2}H_{d-1} = 1 + \frac{1}{2}H_{n-2}$.

Idea of proof. Reveal every V_u with $u \neq o$. In the graph $A \cup \bigcup_{u \neq o} V_u$ the flags at o have degree one and all others degree two, so its components are cycles (exactly the faces avoiding o) and paths joining two distinct flags at o ; the latter define a perfect matching P on the $2d$ flags at o , and the faces through o are the cycles of $P \cup V_o$. By independence V_o is still uniform on $\mathcal{H}_d$. Reveal V_o pair by pair, always querying an unmatched flag x . In $P \cup Q_o$ (with Q_o the pairs revealed so far) the flag x is an endpoint of a path, and exactly one candidate mate, the other endpoint x'' of that path, closes a cycle; every other candidate merges two paths. [*Repair:* the forbidden flag of Lemma 2.2 is the far endpoint of x 's path in $J_o \cup Q_o$, which in general differs from x'' ; if the two coincide the closing probability is 0, otherwise x'' is one of the $2(d-r-1)$ equally likely candidates. In both cases the probability of closing a cycle at this step is at most $1/(2(d-r-1))$. The original writeup relied on the words “at most” without pointing this out.] The forced last pair closes exactly one cycle. Summing over $r = 0, \dots, d-2$ gives $\mathbb{E}F_n^{\text{hit}} \leq \sum_{r=0}^{d-2} \frac{1}{2(d-r-1)} + 1$. [full proof]

4 Faces avoiding the fixed vertex

The trace. Choose a flag $a_0 \in S$, at a vertex $u_0 \neq o$, and let $b_0 = A(a_0)$, at the vertex $w_0 \neq o$. We reveal the face of a_0 one V -pair at a time. The *active flag* is a_0 initially and b_0 is the *terminal*. At each step: reveal the V -mate z of the active flag a_t (a query at the vertex of a_t); if $z = b_0$ the face *closes*; otherwise the new active flag is $a_{t+1} = A(z)$, which is at a different vertex, and the trace *stops* if that vertex is o . Let T be the number of queries until closure or stopping, so $T \leq n(n-1)$. Until closure, the revealed pairs form a single path with endpoints a_t and b_0 , so every query asks for a mate not revealed before and Lemma 2.2 applies at every step; if the trace closes after k queries, the face of a_0 has length k , and if the face of a_0 avoids o the trace does close (Appendix A). Write $r_u(t)$ for the number of pairs revealed at u after t queries (the *load* at u) and $\mathcal{F}_t$ for the information revealed by the first t queries. Let p_o , p_{cl} and p_w be the conditional probabilities, given $\mathcal{F}_t$ and $t < T$, that the next query moves the trace to o , closes the face, or moves it to the vertex $w \neq o$.

Lemma 4.1 (Escape probabilities). *Suppose $t < T$ and the active flag a_t is at $u \neq o$. Then $r_u(t) \leq d-2$ and*

$$p_o = \frac{1}{n-2-r_u(t)} \geq \frac{1}{n-2}, \quad p_w \leq p_o \text{ for every } w \neq o. \quad (4)$$

Consequently, with $q = 1 - \frac{1}{n-2}$,

$$\Pr(T > t) \leq q^t \quad (t \geq 0). \quad (5)$$

Idea of proof. Neither flag of the incidence (u, uo) has been touched, since either would have put the trace at o ; so they form an isolated path of $J_u \cup Q_u$ distinct from the path of a_t , and both are admissible mates, each with probability $1/(2(d-r_u-1))$; two paths also force $r_u \leq d-2$. For $w \neq o$ there are at most two flags at u on the edge uw (simplicity of K_n), each admissible with probability at most $1/(2(d-r_u-1))$. The survival bound follows because the trace terminates with probability at least $p_o \geq 1/(n-2)$ at every step. [full proof]

Lemma 4.2 (Geometric bound on repeated visits). Fix $w \neq o$ and let $N_w(t) = |\{i \leq t : a_i \text{ is at } w\}|$ for $t < T$. For every integer $L \geq 1$,

$$\Pr(N_w(t) \geq L \text{ for some } t < T) \leq 2^{1-L}. \quad (6)$$

Idea of proof. $M_t = 2^{N_w(t)} \mathbf{1}_{\{t < T\}}$ satisfies $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t(1 + p_w - p_o - p_{cl}) \leq M_t$ by (4), the three events being disjoint because b_0 is not at o ; so M is a non-negative supermartingale with $M_0 \leq 2$, and optional stopping at the bounded time $\min\{t : N_w(t) \geq L\} \wedge T$ gives the claim. [\[full proof\]](#)

Good and bad faces. Let

$$R = \lceil 4 \log_2 n \rceil, \quad D = n - 2 - R. \quad (7)$$

[*Repair:* the writeup wrote “for sufficiently large n , $D > 0$ ”. Exactly: $D \geq 1$ if and only if $n \geq 21$ ($n = 20$ gives $D = 0$, $n = 21$ gives $D = 1$, and D is non-decreasing from there on); we therefore assume $n \geq 21$ from now on. The referee estimated the threshold as $n \approx 32$; the exact value was computed during the rewrite. Because R is a ceiling, $2^{-R} \leq n^{-4}$ exactly, which is used in (8).] A face avoiding o is *good* if it uses at most R pairs of V_u at every vertex u , and *bad* otherwise. Let B be the number of bad avoiding faces and G_k the number of good avoiding faces of length k .

Lemma 4.3 (Bad faces). For every $a_0 \in S$, $\Pr(a_0 \text{ lies in a bad avoiding face}) \leq (n-1)2^{-R}$, and

$$\mathbb{E}B \leq (n-1)^2(n-2)2^{-R} \leq n^3 2^{-R} \leq \frac{1}{n}. \quad (8)$$

Idea of proof. A bad avoiding face of a_0 closes at $T = k$ and has at least $R+1$ pairs at some w , i.e. $N_w(k-1) \geq R+1$ with $k-1 < T$; apply (6) with $L = R+1$ and a union bound over the $n-1$ vertices $w \neq o$. Every face of length k has $2k \geq 2$ flags, all in S if it avoids o , so $B = \sum_{a \in S} \mathbf{1}\{a \text{ in a bad avoiding face}\} / (2L(a)) \leq \frac{1}{2} \sum_{a \in S} \mathbf{1}\{\dots\}$, and (3) gives the middle term; the last inequality is $2^{-R} \leq n^{-4}$. [\[full proof\]](#)

Lemma 4.4 (Good faces of a given length). Assume $n \geq 21$. For every $a_0 \in S$ and every $k \geq 2$,

$$\Pr(a_0 \text{ lies in a good avoiding face of length } k) \leq \frac{q^{k-2}}{2D^2}. \quad (9)$$

Idea of proof. [*Repair:* this is the inequality whose justification the referee found compressed to one sentence in the writeup (its equation (11)); the nested-conditioning argument below is the one the referee supplied, and is written out in full in [Appendix A](#).] Let $\mathcal{A} \in \mathcal{F}_{k-2}$ be the event that $T > k-2$ and all loads after $k-2$ queries are at most R ; let $\mathcal{B} \in \mathcal{F}_{k-1}$ be the event that query $k-1$ moves the trace to w_0 ; let $\mathcal{C}$ be the event that query k reveals b_0 . The event in (9) is contained in $\mathcal{A} \cap \mathcal{B} \cap \mathcal{C}$, since loads only grow and the final loads of a good face are at most R . On $\mathcal{A}$ the active vertex u after $k-2$ queries has load at most $R \leq d-2$, so by [Lemma 2.2](#) the next mate is uniform over at least $2D$ flags, at most two of which are on the edge uw_0 : $\Pr(\mathcal{B} \mid \mathcal{F}_{k-2}) \leq 1/D$ on $\mathcal{A}$. On $\mathcal{A} \cap \mathcal{B}$ the active flag a_{k-1} is at w_0 , and the load at w_0 after $k-1$ queries is still at most R , because query $k-1$ was made at $u \neq w_0$ (the active vertex changes at every step); hence $\Pr(\mathcal{C} \mid \mathcal{F}_{k-1}) \leq 1/(2D)$ there. Finally $\Pr(\mathcal{A}) \leq \Pr(T > k-2) \leq q^{k-2}$ by (5). Chaining the three conditional bounds gives (9). [\[full proof\]](#)

Proposition 4.5 (Faces avoiding o). Assume $n \geq 21$. Then

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{(n-1)(n-2)}{2D^2} \sum_{k=3}^{\infty} \frac{q^{k-2}}{k} + \frac{1}{n} = \frac{(n-1)(n-2)}{2D^2} q^{-2} \left(\ln(n-2) - q - \frac{q^2}{2} \right) + \frac{1}{n}, \quad (10)$$

and as $n \rightarrow \infty$,

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4} + O\left(\frac{(\ln n)^2}{n}\right). \quad (11)$$

Idea of proof. Rooting at flags, $\mathbb{E}G_k = \frac{1}{2k} \sum_{a \in S} \Pr(a \text{ lies in a good avoiding face of length } k) \leq \frac{(n-1)(n-2)}{2D^2} q^{k-2}$ by (3) and (9); there are no faces of length ≤ 2 (Lemma 2.3); add $\mathbb{E}B \leq 1/n$. The series is $q^{-2}(-\ln(1-q) - q - q^2/2)$ with $1-q = 1/(n-2)$. Since $R = O(\ln n)$, the prefactor is $\frac{1}{2} + O(\ln n/n)$ and the series is $\ln n - \frac{3}{2} + O(\ln n/n)$. [full proof]

5 Conditioning on non-orientability

Proposition 5.1. *The embedding (π, ε) is orientable with probability $p_n = 2^{-(n-1)(n-2)/2}$, independently of π , and*

$$\mathbb{E}[F_n \mid \text{the embedding is non-orientable}] \leq \frac{\mathbb{E}F_n}{1 - p_n}.$$

In particular (2) holds for the conditional expectation as well.

Idea of proof. An embedding scheme of a connected graph is orientable if and only if its signature is switching-equivalent to the all-positive one [3]; for a connected graph on n vertices there are exactly 2^{n-1} such signatures among the $2^{|E|}$, and $|E(K_n)| = \binom{n}{2}$. Since $F_n \geq 0$, $\mathbb{E}[F_n \mathbf{1}_{\text{non-or.}}] \leq \mathbb{E}F_n$. [full proof]

6 Remarks

Remark 6.1 (Finite n versus the limit). The displayed constant $\frac{1}{4} + \frac{\gamma}{2} \approx 0.5386$ in (2) is a limit; the inequality actually proved for a given n is (1). Evaluating (1): $\mathbb{E}F_{21} \leq 330.6$ (useless: the trivial bound is $n(n-1)/3 = 140$), $\mathbb{E}F_{32} \leq 12.82$, $\mathbb{E}F_{50} \leq 7.98$, $\mathbb{E}F_{100} \leq 6.64$, $\mathbb{E}F_{200} \leq 6.65$, $\mathbb{E}F_{400} \leq 7.01$, $\mathbb{E}F_{800} \leq 7.51$, $\mathbb{E}F_{1600} \leq 8.09$, against $\ln n + 0.5386 = 4.45, 5.14, 5.84, 6.53, 7.22, 7.92$ at $n = 50, 100, 200, 400, 800, 1600$. The referee noted that the writeup’s one-line summary could be misread as a finite- n statement; Theorem 1.2 separates the two.

Remark 6.2 (Simplicity of K_n). Two steps use that K_n has exactly one edge between any two vertices: the inequality $p_w \leq p_o$ in Lemma 4.1, and the bound $\Pr(\mathcal{B} \mid \mathcal{F}_{k-2}) \leq 1/D$ in Lemma 4.4 (“at most two flags at u lead to w_0 ”). The argument does not transfer verbatim to multigraphs, where the neighbouring conjecture of [1, Section 9] that non-orientable random embeddings have no more faces in expectation than orientable ones also lives. The referee asked that this be stated as a hypothesis, which we do in Section 1.

Remark 6.3 (What the referee checked). The referee (Appendix B; scripts in verification_astra/scripts/2211.01032__03/) implemented the model independently and verified: the flag model against the classical orientable permutation model on all 16 ($n = 4$) and 7776 ($n = 5$) rotation systems with an orientable signature (no mismatch; $\mathbb{E}[F(K_5) \mid \text{orientable}] = 1631/648$, matching [2]); the coupling of Lemma 2.1 end-to-end, by sampling V_u from a brute-force enumeration of $\mathcal{H}_{n-1}$ and comparing the law of F with the direct rotation-plus-signature model (total variation 0.0026, 0.0024, 0.0024 at $n = 4, 5, 6$ with $2 \cdot 10^5$ samples, at the Monte Carlo floor); $|\mathcal{H}_d| = 2, 8, 48, 384$ for $d = 2, \dots, 5$; Lemma 2.2 exhaustively over all 384 elements of $\mathcal{H}_5$ and five exposure patterns (support size exactly $2(d-r-1)$, exactly uniform); the exact values $\mathbb{E}F_3 = 3/2$, $\mathbb{E}F_4 = 109/64$, $\mathbb{E}F_n^{\text{hit}}(K_3) = 3/2$ (Lemma 3.1 is tight there) and $\mathbb{E}F_n^{\text{hit}}(K_4) = 101/64 \leq 7/4$; and Proposition 5.1 at $n = 4$ (8 of 64 signatures orientable, $\mathbb{E}[F \mid \text{non-or.}] = 13/8$, $\mathbb{E}[F \mid \text{or.}] = 9/4$). The expansions in Proposition 4.5 were checked symbolically and numerically (at $n = 10^9$ the series is 19.22326 against $\ln n - 1.5 = 19.22327$).

Remark 6.4 (The suspected violation of (9)). A pilot Monte Carlo run with 2000 replicates at $n = 50$ appeared to violate the writeup’s equation (11), i.e. (9), with a worst per-length ratio of 24.7 of empirical to bound. The referee traced this to a single observed good avoiding face of length $k = 278$ (bound 2.03×10^{-5} against an empirical frequency of $1/2000$), and settled the matter by scaling up two hundredfold. Testing the equivalent form $\mathbb{E}G_k \leq C(n)q^{k-2}/k$ with $C(n) = (n-1)(n-2)/(2D^2)$:

n	replicates	R	D	good faces	worst bin	slack $(D/(n-2))^2$	violated
50	400 000	23	25	488 277	0.2698	0.2713	none
100	40 000	27	71	62 144	0.5704	0.5249	none
200	12 000	31	167	22 760	0.7251	0.7114	none

Here the lengths $3 \leq k \leq 1349$ were binned and a bin counts as violated only if the lower 99.9% Poisson confidence limit of its count exceeds the bound. In every bin, at all three values of n , the observed value equals the bound times the slack $(D/(n-2))^2$ that the two last-step factors $1/D$ and $1/(2D)$ predict (their true values being about $1/(n-2)$ and $1/(2(n-2))$), to three significant figures, drifting down at the largest k . The bin $[250, 300)$ at $n = 50$, which contains the alarm, has 99 observations and ratio 0.21. The remaining “worst single- k ratios” (2.8, 16, 146) are each a single count in the far tail with a 99.9% lower limit near 10^{-3} . The referee’s conclusion: (9) is asymptotically sharp but never violated, and the alarm was Monte Carlo noise. Independently of the simulation, the nested-conditioning proof in [Appendix A](#) establishes (9) rigorously.

Remark 6.5 (Sharpness). In the same runs the referee measured $\mathbb{E}F_n^{\text{avoid}} = 1.2207$ ($n = 50$), 1.5536 ($n = 100$) and 1.8967 ($n = 200$), against the limit form $\frac{1}{2} \ln n - \frac{3}{4} = 1.206, 1.553, 1.899$: the measured value matches the limit form to three significant digits at $n = 100$ and to within 0.015 at $n = 50$ (the excess being the positive $O(\ln n/n)$ correction the theorem allows), and to within Monte Carlo error at $n = 200$. The chain of inequalities for the avoiding faces is thus an asymptotic *equality*, while the measured values lie far below the finite- n values 4.75, 3.06, 2.72 of (10). By contrast $\mathbb{E}F_n^{\text{hit}} = 2.9276, 3.2832, 3.6318$ sits a constant ≈ 0.30 below the bound $1 + \frac{1}{2}H_{n-2} = 3.2294, 3.5836, 3.9340$, the slack coming from the “+1” for the forced last pair. Zero bad avoiding faces were observed in any run. Globally, $\mathbb{E}F_n - \ln n$ was measured as 0.2446, 0.2495, 0.2605, 0.2233, 0.4060, 0.2304, 0.2389 at $n = 20, 50, 100, 200, 400, 800, 1600$ (with 20 000 down to 60 replicates; the standard error at $n = 1600$ is 0.33): flat at ≈ 0.24 over two decades, below the proved 0.5386 by almost exactly the 0.30 slack in [Lemma 3.1](#). An independent 400 000-replicate run gives $\mathbb{E}F_{50} = 4.1483$; the orientable random embedding of K_{50} has $\mathbb{E}F \approx 6.98$ [2], consistent with the neighbouring conjecture of [1, Section 9].

Remark 6.6 (The earlier attempt). This is a second attack on the problem. The first (gpt-5.6-so1, verdict “partial”) introduced the flag model and the exposure lemma, and bounded the expected number of faces of length at most αn by $\frac{1}{2} \ln n + O_\alpha(1)$ through a relaxed enumeration that becomes exponentially lossy for long faces; it also claimed a lower bound $\mathbb{E}F_n \geq \frac{1}{4} \ln n - O(1)$ from vertex-simple faces. The present argument keeps only the flag model and the exposure lemma, both re-proved here ([Section 2](#)) and verified by the referee; the long-face obstruction is removed by the fixed-vertex split. The earlier attempt’s lower bound and short-face enumeration are not part of this note and were not refereed.

Remark 6.7 (Novelty and a caveat). The referee enumerated the (small) literature on random 2-cell embeddings by arXiv full-text search and web search in September 2026 and found nothing addressing the non-orientable conjecture, so the result appears to be new; the exposure lemma is a folklore deferred-decision statement already used in the orientable analysis of [1] and is not claimed as new. The referee also observed that a short argument resolving an eighteen-month-old conjecture of a long SODA paper, with an avoiding-face half that is asymptotically tight, is precisely the kind of claim that should receive a human expert’s attention, in particular on [Lemma 4.4](#), before being treated as settled. We repeat that recommendation. Nothing here has been checked by a human.

7 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The argument was found by the language model gpt-6-astra (reasoning effort max) in a single autonomous pass started on 2026-09-16, in the attacks_retry leg of a sweep over open problems catalogued from arXiv (catalog id 2211.01032__03; original writeup in attacks_retry/2211.01032__03/output.md). This was a second attempt: the prompt contained the earlier gpt-5.6-so1 writeup (verdict “partial”) as

context, and no referee report on it. The flag model and the exposure lemma are inherited from that attempt and are not presented as new; the fixed-vertex decomposition, the escape and congestion bounds and the resolution of the conjecture are new to the second attempt (Remark 6.6). (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17: it confirmed the two quotations against the arXiv v3 T_EX source, re-derived every step, implemented the model independently and verified the model reduction, $|\mathcal{H}_d|$, the exposure lemma and the small exact values by enumeration, ran the Monte Carlo experiments of Remarks 6.4 and 6.5, and searched the literature for prior or concurrent work. Its verdict was MINOR_GAPS: no mathematical error, but a compressed justification of the writeup’s equation (11), an unnamed large- n threshold, and the expository points listed below. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The text was reorganised with full proofs in Appendix A. Every deviation from the writeup is one of the following. (a) The one-sentence conditioning argument for equation (11) is replaced by the nested-conditioning proof supplied by the referee, including the observation that query $k - 1$ loads a vertex other than w_0 (Lemma 4.4, Appendix A). (b) The threshold “for sufficiently large n ” is made explicit: $D \geq 1$ iff $n \geq 21$, computed during the rewrite; Theorem 1.2 is stated as a finite- n inequality for $n \geq 21$ plus an asymptotic statement, and the finitely many small n are absorbed in Corollary 1.3 via $F_n \leq n(n - 1)/3$. (c) The role of the ceiling in R , namely $2^{-R} \leq n^{-4}$, is stated where (8) uses it. (d) Simplicity of K_n is stated as a hypothesis and the two places where it is used are named (Remark 6.2). (e) In Lemma 3.1 the distinction between the closing flag (read off $P \cup Q_o$) and the forbidden flag of the exposure lemma (read off $J_o \cup Q_o$) is made explicit. (f) The coupling of Lemma 2.1 is written with an explicit twist bit and relabelling map in place of the writeup’s “immaterial fixed adjustment”. (g) Lemma 2.3 and the trace invariants are spelled out as separate statements. (h) Citations were added for the source paper, the companion note [2] on the orientable model, and two textbooks [3, 4] for embedding schemes and the orientability criterion; the writeup cited nothing. Repairs (a)–(e) are marked where they occur. No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

A Full proofs

Lemma 2.1 (Flag model). *Let (π, ε) be the non-orientable random embedding of K_n . There is a coupling with matchings $(V_u)_{u \in V(K_n)}$, independent and uniform on $\mathcal{H}_d$, such that the faces of (π, ε) correspond bijectively, and length-preservingly, to the cycles of $A \cup V$. Moreover $|\mathcal{H}_d| = 2^{d-1}(d - 1)!$.*

Proof. *The ribbon graph.* Represent each vertex u by a closed disc and each edge $e = uv$ by a band. The band of e is attached to the disc of u along a short arc of its boundary and likewise at v ; the two corners where the band’s two sides meet the boundary of the disc of u are the two flags $(u, v, 0)$ and $(u, v, 1)$, paired by J_u . Fix once and for all, for every incidence, which corner is labelled 0 and which 1, by the orientation of the disc of u (say, 0 is the corner met first when the boundary circle is traversed counter-clockwise). The cyclic order π_u prescribes the order of the d attachment arcs along the boundary circle of the disc of u ; between two consecutive attachments there is an arc of free boundary, which joins the corner 1 of one incidence to the corner 0 of the next. These d free arcs define a perfect matching V_u^π on the flags at u , and $J_u \cup V_u^\pi$ is the boundary circle, a single $2d$ -cycle: $V_u^\pi \in \mathcal{H}_d$. The band of $e = uv$ has two sides; each side joins a corner at u to a corner at v . Whether side s at u is side s or side $1 - s$ at v depends on whether the band is twisted relative to the two disc orientations, so the band matching is

$$(u, v, s) \longleftrightarrow (v, u, s \oplus \tau_e), \quad \tau_e \in \{0, 1\},$$

where $\tau_e = \tau_e(\varepsilon(e))$ is a bijective function of the sign $\varepsilon(e)$; the correspondence between $\varepsilon(e) = -1$ and $\tau_e = 1$ may depend on the convention chosen for the disc orientations, but for each edge it is a fixed bijection $\{\pm 1\} \rightarrow \{0, 1\}$. The boundary of the ribbon graph, i.e. the union of the face boundaries

of the embedding, is the union of the free arcs (the V -edges) and the band sides (the A_τ -edges), so the faces are exactly the cycles of $A_\tau \cup V^\pi$, a face of length k (in edge-sides) being a cycle with k band sides and k free arcs.

Randomising. Since $\varepsilon(e)$ is fair and independent across edges and of π , so is τ_e . Let $(b_{u,e})$ be independent fair bits indexed by the *incidences* (u, e) , independent of π , and set $\tau'_e = b_{u,e} \oplus b_{v,e}$ for $e = uv$. Then $(\tau'_e)_e$ are independent fair bits independent of π , so (π, τ') has the law of (π, τ) and it suffices to prove the lemma for (π, τ') . Define the relabelling $\varphi(u, v, s) = (u, v, s \oplus b_{u,uv})$, a bijection on flags that preserves each J_u and each vertex. It maps the $A_{\tau'}$ -edge $\{(u, v, s), (v, u, s \oplus \tau'_e)\}$ to $\{(u, v, s \oplus b_{u,e}), (v, u, s \oplus \tau'_e \oplus b_{v,e})\} = \{(u, v, s'), (v, u, s')\}$ with $s' = s \oplus b_{u,e}$, an edge of the canonical A ; and it maps V_u^π to $V_u := \varphi(V_u^\pi)$, which lies in $\mathcal{H}_d$ because φ preserves J_u . Hence φ is a bijection from the cycles of $A_{\tau'} \cup V^\pi$ onto the cycles of $A \cup V$, preserving lengths.

Uniformity. V_u is a function of $(\pi_u, (b_{u,e})_{e \ni u})$, which are independent across u ; so the V_u are independent. The pair $(\pi_u, (b_{u,e})_e)$ is uniform over $(d-1)!2^d$ values. Given $W \in \mathcal{H}_d$, the $2d$ -cycle $J_u \cup W$ can be traversed in two directions; a direction determines the cyclic order in which the incidences are met and, for each incidence, which of its two flags is met first, i.e. it determines a pair $(\pi_u, (b_{u,e})_e)$, and the two directions give two different pairs (the flag met first at each incidence becomes the flag met second, so all d bits are complemented; for $d \geq 3$ the cyclic order is reversed as well). Conversely a pair (π_u, b_u) determines the traversal and hence W . So the map $(\pi_u, b_u) \mapsto V_u$ is exactly two-to-one onto $\mathcal{H}_d$: V_u is uniform on $\mathcal{H}_d$ and $|\mathcal{H}_d| = (d-1)!2^d/2 = 2^{d-1}(d-1)!$. $\square$

Lemma 2.2 (Exposure lemma). *Let V_u be uniform on $\mathcal{H}_d$ and let $Q \subseteq V_u$ consist of r pairs. Then $J_u \cup Q$ is a disjoint union of $d-r$ paths whose endpoints are exactly the flags at u not matched by Q . Conditionally on $\{Q \subseteq V_u\}$, for any unmatched flag x :*

- (i) *if $r \leq d-2$, the V_u -mate of x is uniform over the $2(d-r-1)$ unmatched flags other than x and the other endpoint of the path of $J_u \cup Q$ containing x ;*
- (ii) *if $r = d-1$, the mate of x is that other endpoint.*

The same holds when Q is built adaptively, each pair being revealed by querying the mate of a flag chosen as a function of the pairs revealed so far and of randomness independent of V_u .

Proof. $J_u \cup Q \subseteq J_u \cup V_u$ is a subgraph of a cycle, hence a disjoint union of paths (no cycles, since $Q \neq V_u$ when $r \leq d-1$, and a proper subgraph of a cycle contains no cycle). It has $2d$ vertices and $d+r$ edges, so it has $2d - (d+r) = d-r$ components. Every flag has degree one in J_u , so the endpoints of the paths are exactly the flags of degree one, i.e. those not matched by Q ; there are $2(d-r)$ of them.

Conditionally on $\{Q \subseteq V_u\}$, V_u is uniform on $\mathcal{H}_d(Q) = \{W \in \mathcal{H}_d : Q \subseteq W\}$. Contract each path of $J_u \cup Q$ to a single edge between its two endpoints; this produces a set of $2(d-r)$ flags with a perfect matching J' (the contracted paths), and $W \mapsto W \setminus Q$ is a bijection from $\mathcal{H}_d(Q)$ onto $\{W' \text{ perfect matching on the endpoints} : J' \cup W' \text{ is one } 2(d-r)\text{-cycle}\} \cong \mathcal{H}_{d-r}$, because $J_u \cup W$ is a single cycle iff $J' \cup (W \setminus Q)$ is. Hence $|\mathcal{H}_d(Q)| = |\mathcal{H}_{d-r}|$.

Let x be unmatched, with path P_x and other endpoint x' . If $r = d-1$ there is one path and $W \setminus Q = \{xx'\}$ is forced: (ii). If $r \leq d-2$, the mate y of x in $W \in \mathcal{H}_d(Q)$ is an unmatched flag other than x ; $y = x'$ is impossible since $P_x + xx'$ would be a cycle in $J_u \cup W$ shorter than $2d$. For any of the other $2(d-r) - 2 = 2(d-r-1)$ unmatched flags y , adding xy merges P_x with the path $P_y \neq P_x$ into one path, giving $r+1$ exposed pairs and no cycle, so the number of $W \in \mathcal{H}_d(Q)$ with $xy \in W$ is $|\mathcal{H}_d(Q \cup \{xy\})| = |\mathcal{H}_{d-r-1}|$, independent of y . Hence y is uniform over these $2(d-r-1)$ flags: (i).

For adaptive exposure: let the j -th query flag x_j be a function of $(x_1, y_1, \dots, x_{j-1}, y_{j-1})$ and of a random variable Z independent of V_u , where y_i is the revealed mate of x_i . Conditionally on Z and on the revealed pairs $Q = \{x_1 y_1, \dots, x_{j-1} y_{j-1}\}$, the law of V_u is its law given $\{Q \subseteq V_u\}$ (the query sequence is a deterministic function of Q and Z , and Z is independent of V_u), to which the first part applies. $\square$

Lemma 3.1 (Faces through o). $\mathbb{E}F_n^{\text{hit}} \leq 1 + \frac{1}{2}H_{d-1} = 1 + \frac{1}{2}H_{n-2}$.

Proof. Reveal V_u for every $u \neq o$ and consider the graph $\Gamma = A \cup \bigcup_{u \neq o} V_u$ on all flags. A flag not at o has degree two in Γ (one A -edge, one V -edge) and a flag at o has degree one (its A -edge). Hence every component of Γ is a cycle, containing no flag at o , or a path whose two endpoints are flags at o ; these endpoints are distinct, being the two ends of a path. Adding V_o leaves the cycles untouched, and they are exactly the faces avoiding o . Let P be the matching on the $2d$ flags at o pairing the two endpoints of each path; since every flag at o is the endpoint of exactly one path, P is a perfect matching. Contracting each path of Γ to its P -edge is a bijection between the cycles of $A \cup V$ through o and the cycles of $P \cup V_o$, so F_n^{hit} is the number of cycles of $P \cup V_o$. By independence of the V_u , V_o is uniform on $\mathcal{H}_d$ conditionally on P .

Reveal V_o pair by pair: at step $r = 0, 1, \dots, d-1$, with Q_o the r pairs revealed so far, pick any flag x at o unmatched by Q_o and reveal its mate y . In the graph $P \cup Q_o$ every flag has degree one or two, so its components are cycles (whose flags are all matched by Q_o) and paths whose endpoints are the unmatched flags; x is an endpoint of a path $\tilde{P}_x$ with other endpoint $x'' \neq x$. Adding the edge xy creates a cycle iff $y = x''$; otherwise it merges $\tilde{P}_x$ with the path $\tilde{P}_y$. Every cycle of $P \cup V_o$ is created at exactly one step (the step at which its last V_o -pair is revealed), so $F_n^{\text{hit}} = \sum_{r=0}^{d-1} \mathbf{1}\{\text{step } r \text{ creates a cycle}\}$.

At step $r \leq d-2$, by [Lemma 2.2\(i\)](#) (applied adaptively, with P as the independent randomness Z), y is uniform over $2(d-r-1)$ flags, namely the unmatched flags other than x and the far endpoint x' of the path of x in $J_o \cup Q_o$. The flags x' and x'' are defined by different path structures and need not coincide. If $x'' = x'$, the event $y = x''$ has probability 0; if $x'' \neq x'$, then x'' is an unmatched flag other than x and x' , so it is one of the $2(d-r-1)$ equally likely candidates and the event $y = x''$ has probability exactly $1/(2(d-r-1))$. In both cases $\Pr(\text{step } r \text{ creates a cycle} \mid \text{past}) \leq 1/(2(d-r-1))$. At step $r = d-1$ there is one unmatched pair $\{x, x''\}$ left and the forced mate is x'' , creating exactly one cycle. Therefore

$$\mathbb{E}F_n^{\text{hit}} \leq \sum_{r=0}^{d-2} \frac{1}{2(d-r-1)} + 1 = 1 + \frac{1}{2} \sum_{j=1}^{d-1} \frac{1}{j} = 1 + \frac{1}{2}H_{d-1} = 1 + \frac{1}{2}H_{n-2}. \quad \square$$

Lemma A.1 (Trace invariants). *In the trace of [Section 4](#), as long as the face has not closed: (a) the revealed V -pairs together with the A -edges a_0b_0 and $z_i a_i$ form a single path from b_0 to the active flag a_t , all of whose flags are distinct; (b) the mate of a_t has not been revealed, so [Lemma 2.2](#) applies to the query at a_t , the revealed pairs at that vertex being those on the path; (c) if the query returns $z \neq b_0$ then z is not on the path. Moreover, if the trace closes after k queries the face of a_0 has length k , and if the face of a_0 avoids o the trace closes.*

Proof. By induction on t . Initially the path is $b_0 A a_0$. Suppose (a) holds after t queries. The flag a_t is an endpoint of the path and its only path-edge is an A -edge, so its V -mate has not been revealed: (b). The revealed pairs at the vertex of a_t are V -edges of the path (the trace reveals nothing else), and every revealed pair is on the path, so [Lemma 2.2](#) applies with Q equal to the path's V -pairs at that vertex; adaptivity is legitimate because the query flag is a function of the pairs revealed so far and of the other, independent, local matchings. Let z be the revealed mate. If z were an interior flag of the path it would already have V -degree one on the path, which is impossible as V is a matching; $z \neq a_t$; so z is on the path only if $z = b_0$: (c). If $z \neq b_0$, then $a_{t+1} = A(z)$ is not on the path either (its A -partner z is not), and the path extends by $a_t V z A a_{t+1}$: (a) for $t+1$. If $z = b_0$ after k queries, the path closes into a cycle of $A \cup V$ with k V -edges and k A -edges, which is the face of a_0 , of length k . Finally, the face of a_0 is the component of a_0 in $A \cup V$; the trace follows it flag by flag, so if that face contains no flag at o the trace never stops at o and, the face being finite, closes. $\square$

Lemma 4.1 (Escape probabilities). *Suppose $t < T$ and the active flag a_t is at $u \neq o$. Then $r_u(t) \leq d-2$ and*

$$p_o = \frac{1}{n-2-r_u(t)} \geq \frac{1}{n-2}, \quad p_w \leq p_o \text{ for every } w \neq o. \quad (4)$$

Consequently, with $q = 1 - \frac{1}{n-2}$,

$$\Pr(T > t) \leq q^t \quad (t \geq 0). \quad (5)$$

Proof. Let $t < T$ with a_t at $u \neq o$, and let Q_u be the pairs revealed at u , $r_u = r_u(t) = |Q_u|$. Consider the two flags $(u, o, 0)$, $(u, o, 1)$ at u on the edge uo . Neither is on the trace path: if $(u, o, s) = a_i$ for some $i \geq 1$, then $z_i = A(a_i) = (o, u, s)$ would be at o , but z_i is at the vertex of a_{i-1} , so a_{i-1} would be at o , contradicting $t < T$; $(u, o, s) = a_0$ or b_0 is impossible since $a_0 \in S$; and if $(u, o, s) = z_i$ for some i , then $a_i = A(z_i) = (o, u, s)$ is at o and the trace stopped after $i \leq t$ queries, again contradicting $t < T$. Hence the pair $\{(u, o, 0), (u, o, 1)\}$ is a J_u -edge none of whose flags is matched by Q_u : an isolated path of $J_u \cup Q_u$, distinct from the path containing a_t . Two distinct paths give $d - r_u \geq 2$, so $r_u \leq d - 2$ and Lemma 2.2(i) applies: the mate z of a_t is uniform over $2(d - r_u - 1)$ flags, which include $(u, o, 0)$ and $(u, o, 1)$ (unmatched, different from a_t , and not the far endpoint of a_t 's path). The trace moves to o iff $z \in \{(u, o, 0), (u, o, 1)\}$ (then $a_{t+1} = (o, u, s)$; note $z \neq b_0$ as $b_0 \notin \{(u, o, s)\}$), so

$$p_o = \frac{2}{2(d - r_u - 1)} = \frac{1}{d - 1 - r_u} = \frac{1}{n - 2 - r_u} \geq \frac{1}{n - 2}.$$

For $w \neq o$, the trace moves to w iff $z \in \{(u, w, 0), (u, w, 1)\}$ and $z \neq b_0$; these are at most two flags, because K_n has exactly one edge between u and w , and each has probability at most $1/(2(d - r_u - 1))$ of being z (exactly that if admissible, 0 otherwise). Hence $p_w \leq 2/(2(d - r_u - 1)) = p_o$.

For (5): on $\{T > t\}$ the active flag is at some $u \neq o$, and the next query terminates the trace (moves to o or closes) with conditional probability at least $p_o \geq 1/(n - 2)$, so $\Pr(T > t + 1 \mid \mathcal{F}_t) \leq q \mathbf{1}_{\{T > t\}}$. Taking expectations and inducting, $\Pr(T > t) \leq q^t$. $\square$

Lemma 4.2 (Geometric bound on repeated visits). *Fix $w \neq o$ and let $N_w(t) = |\{i \leq t : a_i \text{ is at } w\}|$ for $t < T$. For every integer $L \geq 1$,*

$$\Pr(N_w(t) \geq L \text{ for some } t < T) \leq 2^{1-L}. \quad (6)$$

Proof. Set $M_t = 2^{N_w(t)}$ if $t < T$ and $M_t = 0$ if $t \geq T$. On $\{t \geq T\}$, $M_{t+1} = M_t = 0$. On $\{t < T\}$, the query $t + 1$ has three mutually exclusive terminal outcomes and one continuing outcome: it moves the trace to o (probability p_o ; then $T = t + 1$ and $M_{t+1} = 0$), it closes the face (probability p_{cl} ; then $M_{t+1} = 0$), it moves the trace to w (probability p_w ; then $N_w(t + 1) = N_w(t) + 1$ and $M_{t+1} = 2M_t$), or it moves the trace to some vertex other than o and w (then $M_{t+1} = M_t$). Moving to o and closing are disjoint because closing means $z = b_0$, which is not at o , while moving to o means z is on the edge uo . Hence

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t(2p_w + (1 - p_o - p_{cl} - p_w)) = M_t(1 + p_w - p_o - p_{cl}) \leq M_t$$

by (4). So (M_t) is a non-negative supermartingale, with $M_0 = 2^{N_w(0)} \leq 2$ ($N_w(0) = 1$ if a_0 is at w , else 0). Let $\tau = \min\{t : N_w(t) \geq L \text{ or } t \geq T\}$; this is a stopping time bounded by $n(n - 1) + 1$, since each query reveals a new V -pair and there are $n(n - 1)$ of them. By optional stopping $\mathbb{E}M_\tau \leq \mathbb{E}M_0 \leq 2$. On the event $\{N_w(t) \geq L \text{ for some } t < T\}$, at the first such t we have $t < T$ and $\tau = t$, so $M_\tau = 2^{N_w(t)} \geq 2^L$. By Markov's inequality the event has probability at most $2 \cdot 2^{-L} = 2^{1-L}$. $\square$

Lemma 4.3 (Bad faces). *For every $a_0 \in S$, $\Pr(a_0 \text{ lies in a bad avoiding face}) \leq (n - 1)2^{-R}$, and*

$$\mathbb{E}B \leq (n - 1)^2(n - 2)2^{-R} \leq n^3 2^{-R} \leq \frac{1}{n}. \quad (8)$$

Proof. Suppose the face of a_0 avoids o and is bad. By Lemma A.1 the trace closes at some $T = k$, and the V -pairs of the face are $\{a_i z_{i+1}\}$, $0 \leq i \leq k - 1$, the pair $a_i z_{i+1}$ lying at the vertex of a_i . Badness means that for some vertex w at least $R + 1$ of these pairs are at w , i.e. $N_w(k - 1) = |\{i \leq k - 1 :$

a_i at $w\}$ $\geq R + 1$, and $k - 1 < T$. Since the face avoids o , $w \neq o$. By Lemma 4.2 with $L = R + 1$ and a union bound over the $n - 1$ vertices $w \neq o$,

$$\Pr(a_0 \text{ lies in a bad avoiding face}) \leq (n - 1)2^{-R}.$$

A face of length k has $2k$ flags, all in S if it avoids o , so counting each bad avoiding face once through its flags,

$$B = \sum_{a \in S} \frac{\mathbf{1}\{a \text{ lies in a bad avoiding face}\}}{2L(a)} \leq \frac{1}{2} \sum_{a \in S} \mathbf{1}\{a \text{ lies in a bad avoiding face}\},$$

where $L(a) \geq 1$ is the length of the face of a . Taking expectations and using (3),

$$\mathbb{E}B \leq \frac{1}{2} \cdot 2(n - 1)(n - 2) \cdot (n - 1)2^{-R} = (n - 1)^2(n - 2)2^{-R} \leq n^3 2^{-R} \leq n^3 n^{-4} = \frac{1}{n},$$

where $2^{-R} \leq n^{-4}$ because $R = \lceil 4 \log_2 n \rceil \geq 4 \log_2 n$. $\square$

Lemma 4.4 (Good faces of a given length). *Assume $n \geq 21$. For every $a_0 \in S$ and every $k \geq 2$,*

$$\Pr(a_0 \text{ lies in a good avoiding face of length } k) \leq \frac{q^{k-2}}{2D^2}. \quad (9)$$

Proof. Fix $a_0 \in S$ and $k \geq 2$. Define the events

$$\begin{aligned} \mathcal{A} &= \{T > k - 2\} \cap \{r_u(k - 2) \leq R \text{ for every vertex } u\} \in \mathcal{F}_{k-2}, \\ \mathcal{B} &= \{T > k - 1\} \cap \{a_{k-1} \text{ is at } w_0\} \in \mathcal{F}_{k-1}, \\ \mathcal{C} &= \{\text{query } k \text{ reveals } z = b_0\}. \end{aligned}$$

Let $\mathcal{G}_k$ be the event that a_0 lies in a good avoiding face of length k . On $\mathcal{G}_k$ the trace closes at $T = k$ (Lemma A.1), so $T > k - 2$ and $T > k - 1$; the loads are non-decreasing in t and the final loads $r_u(k)$ are the numbers of V -pairs of the face at u , at most R by goodness, so $r_u(k - 2) \leq R$ for all u ; query k reveals b_0 , so $\mathcal{C}$ holds; and a_{k-1} is the flag whose mate is b_0 , hence is at the vertex w_0 of b_0 . Thus $\mathcal{G}_k \subseteq \mathcal{A} \cap \mathcal{B} \cap \mathcal{C}$, and it suffices to bound $\Pr(\mathcal{A} \cap \mathcal{B} \cap \mathcal{C})$.

Step 1: $\Pr(\mathcal{C} \mid \mathcal{F}_{k-1}) \leq \frac{1}{2D}$ on $\mathcal{A} \cap \mathcal{B}$. On $\mathcal{A} \cap \mathcal{B}$ the active flag a_{k-1} is at w_0 . Query $k - 1$ was made at the vertex u of a_{k-2} , and $a_{k-1} = A(z_{k-1})$ is at the other end of the edge on which z_{k-1} lies, so $u \neq w_0$: the active vertex changes at every step. Hence query $k - 1$ did not change the load at w_0 , and $r_{w_0}(k - 1) = r_{w_0}(k - 2) \leq R$ by $\mathcal{A}$. As $R \leq d - 2$ (this is $D \geq 1$), Lemma 2.2(i) applies at a_{k-1} (Lemma A.1(b)): the revealed mate is uniform over $2(d - r_{w_0}(k - 1) - 1) \geq 2(d - R - 1) = 2D$ flags, of which b_0 is at most one. So $\Pr(\mathcal{C} \mid \mathcal{F}_{k-1}) \leq 1/(2D)$ on $\mathcal{A} \cap \mathcal{B}$.

Step 2: $\Pr(\mathcal{B} \mid \mathcal{F}_{k-2}) \leq \frac{1}{D}$ on $\mathcal{A}$. On $\mathcal{A}$ the trace is alive after $k - 2$ queries, with active flag a_{k-2} at some $u \neq o$ and $r_u(k - 2) \leq R \leq d - 2$. By Lemma 2.2(i) the mate z_{k-1} is uniform over at least $2D$ flags. The event $\mathcal{B}$ requires $z_{k-1} \in \{(u, w_0, 0), (u, w_0, 1)\}$ (so that $a_{k-1} = A(z_{k-1})$ is at w_0), at most two flags since K_n is simple (and no flags at all if $u = w_0$). So $\Pr(\mathcal{B} \mid \mathcal{F}_{k-2}) \leq 2/(2D) = 1/D$ on $\mathcal{A}$.

Step 3: $\Pr(\mathcal{A}) \leq q^{k-2}$. $\mathcal{A} \subseteq \{T > k - 2\}$ and (5).

Chaining. Since $\mathcal{A} \in \mathcal{F}_{k-2}$ and $\mathcal{B} \in \mathcal{F}_{k-1}$,

$$\Pr(\mathcal{A} \cap \mathcal{B} \cap \mathcal{C}) = \mathbb{E}[\mathbf{1}_{\mathcal{A}} \mathbf{1}_{\mathcal{B}} \Pr(\mathcal{C} \mid \mathcal{F}_{k-1})] \leq \frac{1}{2D} \mathbb{E}[\mathbf{1}_{\mathcal{A}} \Pr(\mathcal{B} \mid \mathcal{F}_{k-2})] \leq \frac{1}{2D} \cdot \frac{1}{D} \Pr(\mathcal{A}) \leq \frac{q^{k-2}}{2D^2}.$$

Note that Steps 1 and 2 bound conditional probabilities given the revealed history on events measurable with respect to that history; nothing is conditioned on the eventual goodness of the face, which is not $\mathcal{F}_{k-1}$ -measurable. This is the argument the original writeup summarised as “discard continuations whose histories cease to satisfy the load restriction”. $\square$

Proposition 4.5 (Faces avoiding o). *Assume $n \geq 21$. Then*

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{(n-1)(n-2)}{2D^2} \sum_{k=3}^{\infty} \frac{q^{k-2}}{k} + \frac{1}{n} = \frac{(n-1)(n-2)}{2D^2} q^{-2} \left(\ln(n-2) - q - \frac{q^2}{2} \right) + \frac{1}{n}, \quad (10)$$

and as $n \rightarrow \infty$,

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4} + O\left(\frac{(\ln n)^2}{n}\right). \quad (11)$$

Proof. Every good avoiding face of length k has $2k$ flags, all in S , so

$$G_k = \frac{1}{2k} \sum_{a \in S} \mathbf{1}\{a \text{ lies in a good avoiding face of length } k\}$$

and, by (3) and Lemma 4.4,

$$\mathbb{E}G_k \leq \frac{2(n-1)(n-2)}{2k} \cdot \frac{q^{k-2}}{2D^2} = \frac{(n-1)(n-2)}{2D^2} \frac{q^{k-2}}{k} \quad (k \geq 2).$$

By Lemma 2.3, $G_k = 0$ for $k \leq 2$. Every avoiding face is good or bad, so $F_n^{\text{avoid}} = \sum_{k \geq 3} G_k + B$ and Lemma 4.3 gives the first inequality in (10). For the series, with $0 < q < 1$,

$$\sum_{k=3}^{\infty} \frac{q^{k-2}}{k} = q^{-2} \sum_{k=3}^{\infty} \frac{q^k}{k} = q^{-2} \left(-\ln(1-q) - q - \frac{q^2}{2} \right) = q^{-2} \left(\ln(n-2) - q - \frac{q^2}{2} \right),$$

since $1-q = 1/(n-2)$.

For (11): $q^{-2} = (1 - \frac{1}{n-2})^{-2} = 1 + O(1/n)$ and $\ln(n-2) = \ln n + O(1/n)$, so the series equals $\ln n - 1 - \frac{1}{2} + O(\ln n/n) = \ln n - \frac{3}{2} + O(\ln n/n)$. Since $R = O(\ln n)$, $\frac{(n-1)(n-2)}{2(n-2-R)^2} = \frac{1}{2} \cdot \frac{n-1}{n-2} \cdot (1 - \frac{R}{n-2})^{-2} = \frac{1}{2} + O(\ln n/n)$. Multiplying, $\mathbb{E}F_n^{\text{avoid}} \leq (\frac{1}{2} + O(\frac{\ln n}{n}))(\ln n - \frac{3}{2} + O(\frac{\ln n}{n})) + \frac{1}{n} = \frac{1}{2} \ln n - \frac{3}{4} + O(\frac{(\ln n)^2}{n})$. $\square$

Theorem 1.2 (Main theorem). *Let $R = \lceil 4 \log_2 n \rceil$, $D = n - 2 - R$ and $q = 1 - \frac{1}{n-2}$. For every $n \geq 21$ (so that $D \geq 1$),*

$$\mathbb{E}F_n \leq 1 + \frac{1}{2}H_{n-2} + \frac{(n-1)(n-2)}{2D^2} q^{-2} \left(\ln(n-2) - q - \frac{q^2}{2} \right) + \frac{1}{n}. \quad (1)$$

Consequently, as $n \rightarrow \infty$,

$$\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right). \quad (2)$$

Proof. Let $n \geq 21$, so $D \geq 1$ (Eq. (7) and the remark after it). By Lemma 2.1 we may compute $\mathbb{E}F_n$ in the flag model; fix a vertex o and write $F_n = F_n^{\text{hit}} + F_n^{\text{avoid}}$. Lemma 3.1 gives $\mathbb{E}F_n^{\text{hit}} \leq 1 + \frac{1}{2}H_{n-2}$ and Proposition 4.5 gives (10); adding them is (1). For (2), $H_{n-2} = \ln(n-2) + \gamma + O(1/n) = \ln n + \gamma + O(1/n)$, so $\mathbb{E}F_n^{\text{hit}} \leq \frac{1}{2} \ln n + 1 + \frac{\gamma}{2} + O(1/n)$; adding (11),

$$\mathbb{E}F_n \leq \ln n + 1 - \frac{3}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right) = \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right). \quad \square$$

Proposition 5.1. *The embedding (π, ε) is orientable with probability $p_n = 2^{-(n-1)(n-2)/2}$, independently of π , and*

$$\mathbb{E}[F_n \mid \text{the embedding is non-orientable}] \leq \frac{\mathbb{E}F_n}{1 - p_n}.$$

In particular (2) holds for the conditional expectation as well.

Proof. An embedding scheme (π, ε) of a connected graph G is orientable if and only if the signature ε can be made all-positive by *switchings* (replacing ε by $\varepsilon \cdot \sigma(u)\sigma(v)$ on each edge uv , for some $\sigma: V(G) \rightarrow \{\pm 1\}$, which corresponds to reversing the rotations at the vertices with $\sigma = -1$), equivalently iff every cycle of G carries an even number of negative edges [3, 4]. The orientable signatures are therefore $\{e = uv \mapsto \sigma(u)\sigma(v)\}$ for $\sigma \in \{\pm 1\}^{V(G)}$; for connected G the map $\sigma \mapsto \varepsilon_\sigma$ is exactly two-to-one (σ and $-\sigma$ give the same ε , and $\varepsilon_\sigma = \varepsilon_{\sigma'}$ forces $\sigma\sigma'$ constant on the connected graph), so there are 2^{n-1} of them among the $2^{|E(G)|}$ signatures. For K_n , $|E| = \binom{n}{2}$ and the signature is uniform and independent of π , so

$$p_n = \Pr(\text{orientable}) = 2^{n-1} / 2^{\binom{n}{2}} = 2^{-(n-1)(n-2)/2},$$

independently of π . Since $F_n \geq 0$, $\mathbb{E}[F_n \mid \text{non-or.}] = \mathbb{E}[F_n \mathbf{1}_{\text{non-or.}}] / (1 - p_n) \leq \mathbb{E}F_n / (1 - p_n)$, and $1/(1 - p_n) = 1 + O(2^{-n^2/3})$ does not affect (2). $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its section and equation numbers refer to that writeup, not to this note: the writeup’s (1) is Lemma 2.2(i), its (2) is Lemma 3.1, (3)–(4) are (4), (5) is (5), (6) is (6), (7) is (7), (8) is (3), (9)–(10) are Lemma 4.3 and (8), (11) is (9), (12)–(15) are Proposition 4.5, and its §2, §3, §4 are our Sections 3 and 4. The items it lists under “Caveats” are the ones repaired here (see Section 7). The scripts it mentions are in `verification_astra/scripts/2211.01032__03/`. The text is unchanged; only line-breaking hints were inserted for typesetting. Treat it as machine output, not as an authority.

B.1 Interpretation

Source statement, verbatim. I downloaded the arXiv v3 TeX source (<https://arxiv.org/e-print/2211.01032v3>, file RE-arxiv3.tex, line 2604, third conjecture of the closing “Open Problems” discussion):

The expected number of faces in a non-orientable random embedding of the complete graph K_n is at most $\ln(n) + O(1)$.

preceded (line 2601) by the definition of the model:

One natural way to define a random embedding of a graph on a non-orientable surface is to randomly choose a rotation system, and randomly choose a signature for all the edges in the graph, with probability $1/2$ of being either sign. From data, we expect a similar result to hold for non-orientable random embeddings of K_n under this definition.

This matches the catalog text exactly, and it is exactly the model the writeup uses: uniform local rotation at each vertex (a uniform cyclic order, $(d-1)!$ choices, $d = n-1$) together with i.i.d. fair edge signs, **unconditioned**. The source does not condition on the resulting surface being non-orientable; the writeup nevertheless disposes of that alternative reading in its last section (the orientable signatures are the 2^{n-1} coboundaries, so they have probability $2^{-(n-1)(n-2)/2}$ and conditioning changes nothing). So both readings are covered and the interpretation is the intended one; a reasonable author of the conjecture would regard it as resolved by this theorem. No interpretation gaming: the writeup proves the upper bound the conjecture asserts, in the stronger explicit form $\ln n + \frac{1}{4} + \frac{7}{2} + o(1)$.

Retry-leg discipline. The prompt contains an earlier gpt-5.6-`so1` attempt (verdict `partial`) and no referee report on it. The new writeup inherits only the flag/ribbon-graph representation and the exposure lemma, and it re-derives both (§1); I checked both from scratch rather than crediting them. It explicitly discards the earlier short-face enumeration, and it does **not** reproduce the earlier attempt’s actual weakness (the abstract-stub relaxation of its §5, which is exponentially lossy for long faces). The route is genuinely different: a fixed-vertex split plus a per-trace congestion supermartingale.

B.2 Step-by-step findings

step	label	note
§1 S1. Flag model: $2d$ flags/vertex; matchings J_u, A, V_u ; boundary components of $A \cup V$ are the faces; a length- k face has k A -edges, k V -edges, $2k$ flags; $\sum_f f = n(n-1)$	VALID	Implemented independently (model.py). Cross-checked against the classical orientable permutation model: for every one of the 16 ($n=4$) and 7776 ($n=5$) rotation systems with an orientable signature the flag count equals $\#cycles(\pi) - 0$ mismatches — and gives $\mathbb{E}[F(K_5) \mid \text{orientable}] = 1631/648$, matching the value independently obtained in the sibling report verification/2211.01032__02.md. Exact non-orientable-model values: $\mathbb{E}F_3 = 3/2$, $\mathbb{E}F_4 = 109/64$.
§1 S2. $\mathcal{H}_d$; the V_u may be taken i.i.d. uniform on $\mathcal{H}_d$ with A canonical (signature absorbed into per- incidence bits $b_{u,e}$, $\varepsilon_e = b_{u,e} \oplus b_{v,e}$)	VALID	The bits are per incidence (two per edge), so the ε_e really are i.i.d. fair — this is the point at which a per-vertex switching would have failed (coboundaries only). Verified end-to-end (hmodel_equiv.py): sampling V_u uniformly from a brute-force enumeration of $\mathcal{H}_{n-1}$ with A canonical reproduces the law of F from the direct rotation+signature model — total variation 0.0026, 0.0024, 0.0024 for $n=4, 5, 6$ at $2 \cdot 10^5$ samples, i.e. at the Monte-Carlo floor $\approx 1/\sqrt{\text{reps}} = 0.0022$. The hand-waved “immaterial fixed adjustment if the opposite sign convention is used” is thereby confirmed rather than taken on trust.
§1 S3. $ \mathcal{H}_d = 2^{d-1}(d-1)!$	VALID	Brute force: $ \mathcal{H}_d = 2, 8, 48, 384$ for $d=2, 3, 4, 5$, matching exactly.
§1 S4. Exposure lemma (1): after $r \leq d-2$ exposed pairs the mate of a free flag x is uniform on the $2(d-r-1)$ free flags other than x and the far endpoint of x 's $J_u \cup Q_u$ path; forced when $r = d-1$; valid under adaptive exposure	VALID	Exhaustively verified for $d=5$ over all 384 elements of $\mathcal{H}_5$ and five exposure patterns $r=0, \dots, 4$: support size exactly $2(d-r-1)$ and exactly uniform in every case ($r=4$ is the forced case the writeup flags). Adaptivity is legitimate because the V_u are independent across vertices and the query flag is a function of already-revealed pairs.
§2 S5. $F_n = F^{\text{hit}} + F^{\text{avoid}}$; exposing all V_u , $u \neq o$, leaves cycles (= avoiding faces) and paths whose endpoints are the $2d$ flags at o , inducing a perfect matching P ; faces meeting $o =$ components of $P \cup V_o$	VALID	Flags at o have degree 1 in the exposed graph (A only), all others degree 2; A never joins two flags at the same vertex, so each path has two distinct o -endpoints and P is a perfect matching. V_o stays uniform by independence.
§2 S6. Pair-by-pair exposure of V_o ; at most one mate closes a component; probability $\leq 1/(2(d-r-1))$; the forced last pair closes exactly one $\Rightarrow$ (2) $\mathbb{E}F^{\text{hit}} \leq 1 + \frac{1}{2}H_{n-2}$	VALID	Re-derived: in $P \cup Q_o$ a free flag is an endpoint of its path, so only the far endpoint of <i>that</i> path closes a cycle; every other choice merges two paths. Subtle point the writeup handles only through the word “at most”: the <i>closing</i> flag is read off $P \cup Q_o$ while the <i>forbidden</i> flag of (1) is read off $J_o \cup Q_o$ — these differ, but the closing flag then has probability either $1/(2(d-r-1))$ or 0. Sum $\sum_{r=0}^{d-2} \frac{1}{2(d-r-1)} + 1 = 1 + H_{d-1}/2$, $d-1 = n-2$. Verified: exact $\mathbb{E}F^{\text{hit}}(K_3) = 3/2 =$ the bound (tight), $\mathbb{E}F^{\text{hit}}(K_4) = 101/64 \leq 7/4$ (and $101/64 + 1/8 = 109/64 = \mathbb{E}F_4$); Monte Carlo $2.9276 \leq 3.2294$ ($n=50$), $3.2832 \leq 3.5836$ ($n=100$), $3.6318 \leq 3.9340$ ($n=200$) — slack a constant ≈ 0.30 throughout, never violated.
§3 S7. Trace process; “every query is to a previously unexposed mate”; closure after k queries $\Leftrightarrow$ face length k	VALID	The exposed pairs form one path with endpoints (active flag, b_0); a repeated flag would force $z_t = b_0$, i.e. closure. Counting: k A -edges, k V -pairs, $2k$ flags.

step	label	note
§3 S8. (3) $p_o = 1/(n-2-r_u)$ exactly, and the forced-last-pair case cannot occur while avoiding o	VALID	Both flags of the incidence (u, uo) are untouched (either would have put the trace at o), so they form an isolated J_u -path distinct from the active flag's path; both are admissible, giving $2/(2(d-r_u-1))$. Two unfinished paths $\Rightarrow r_u \leq d-2$.
§3 S9. (4) $p_w \leq p_o$ for $w \neq o$	VALID	At most two flags at u have A -partners at w (simple graph — this is where K_n being simple is used); each has probability $\leq 1/(2(d-r_u-1))$.
§3 S10. (5) $\Pr(T > t) \leq q^t$, $q = 1 - 1/(n-2)$	VALID	Immediate from $p_o \geq 1/(n-2)$; survival also has to avoid closure, so the bound is conservative. Monte-Carlo check in <code>trace_test.py</code> shows no t with a significant excess.
§3 S11. (6) $M_t = 2^{N_w(t)} \mathbf{1}_{t < T}$ is a supermartingale, $M_0 \leq 2$, optional stopping $\Rightarrow \Pr(N_w \geq L) \leq 2^{1-L}$	VALID	$\mathbb{E}[M_{t+1} \mathcal{F}_t] = M_t(1 - p_o - p_{cl} - p_w + 2p_w) \leq M_t$ by (4); the three events are disjoint because b_0 is not at o (the start edge avoids o). Bounded stopping time ($\leq n(n-1)$).
§4 S12. (7) $R = \lceil 4 \log_2 n \rceil$, $D = n - 2 - R > 0$; good/bad; (8) $ S = 4 \binom{n-1}{2} = 2(n-1)(n-2)$	VALID	Arithmetic correct. $D > 0$ needs $n \gtrsim 33$ ($n = 20 \Rightarrow D = 0$; $n = 50 \Rightarrow D = 25$); see Caveats.
§4.1 S13. (9) $\Pr(\text{start flag in a bad avoiding face}) \leq (n-1)2^{-R}$; (10) $\mathbb{E}B \leq (n-1)^2(n-2)2^{-R} \leq n^3 2^{-R} \leq 1/n$	VALID	Load at w = number of active visits to w , all occurring at times $t \leq T-1 < T$, so (6) with $L = R+1$ applies; union over the $n-1$ vertices $\neq o$. $2^{-R} \leq n^{-4}$ holds exactly because of the ceiling. Empirically vacuous but never violated: zero bad avoiding faces in 400 000 ($n = 50$), 40 000 ($n = 100$) and 12 000 ($n = 200$) replicates.
§4.2 S14. (11) $\Pr(\text{start flag in a good avoiding face of length } k) \leq q^{k-2}/(2D^2)$	VALID (justification compressed — see Caveats)	This was the flagged item. It is true, and rigorously so, but the writeup's whole argument for the conditioning is the sentence “discard continuations whose histories cease to satisfy the load restriction”. The missing derivation: let $A_{k-2} = \{T > k-2\} \cap \{\text{all loads} \leq R\} \in \mathcal{F}_{k-2}$, $B = \{\text{query } k-1 \text{ moves to } w_0\}$, $C = \{\text{query } k \text{ reveals } b_0\}$; then $\Pr(A_{k-2} \cap B \cap C) = \mathbb{E}[\mathbf{1}_{A_{k-2}} \mathbf{1}_B \Pr(C \mathcal{F}_{k-1})]$, and on $A_{k-2} \cap B$ one needs $r_{w_0} \leq R$ at time $k-1$ — which holds because query $k-1$ exposes a pair at $u \neq w_0$ (the active vertex changes at every step, A always crossing an edge), so the load at w_0 is unchanged. Hence $\Pr \leq q^{k-2} \cdot \frac{1}{D} \cdot \frac{1}{2D}$. Numerically decisive (below): no violation at any k at $n = 50, 100, 200$.
§4.3 S15. No faces of length 1 or 2 for $n \geq 3$	VALID	A length-2 face is a 4-cycle $x_1-A-x_2-V-x_3-A-x_4-V-x_1$, forcing both A -edges onto the single edge uv of K_n and hence a V -pair equal to a J -edge — a proper cycle in $J \cup V$, excluded from $\mathcal{H}_d$ for $d > 1$.
§4.3 S16. (12) $\mathbb{E}G_k \leq \frac{(n-1)(n-2)}{2D^2} q^{k-2}$; (13) $\sum_{k \geq 3} \frac{q^{k-2}}{k} = q^{-2}(\ln(n-2) - q - \frac{q^2}{2}) = \ln n - \frac{3}{2} + O(\frac{\ln n}{n})$; (14) $\frac{(n-1)(n-2)}{2D^2} = \frac{1}{2} + O(\frac{\ln n}{n})$; (15) $\mathbb{E}F^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4} + O(\frac{(\ln n)^2}{n})$	VALID	Rooting factor: a length- k face carries $2k$ flags, all in S when it avoids o . All three expansions re-derived symbolically and checked numerically (<code>arith.py</code>): the closed form of (13) agrees with the truncated series to 10^{-9} , and (13), (14), (15) converge to the stated limits ($n = 10^9$: series 19.22326 vs $\ln n - 1.5 = 19.22327$; factor 0.500000; product 9.611635 vs $\frac{1}{2} \ln n - \frac{3}{4} = 9.611633$).
§5 S17. $\mathbb{E}F^{\text{hit}} \leq \frac{1}{2} \ln n + 1 + \frac{\gamma}{2} + O(1/n)$; total $\leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O(\frac{(\ln n)^2}{n})$	VALID	$1 + H_{n-2}/2 = 1 + \frac{1}{2}(\ln(n-2) + \gamma) + O(1/n)$; addition correct; $1 - \frac{3}{4} = \frac{1}{4}$.

step	label	note
§5 S18. Non-orientability conditioning: 2^{n-1} orientable signatures, $p_n = 2^{-(n-1)(n-2)/2}$, $\mathbb{E}[F \mid \text{non-or.}] \leq \mathbb{E}F/(1 - p_n)$	VALID	Orientability $\Leftrightarrow$ the signature lies in the $(n-1)$ -dimensional cut space, independent of the rotation. Verified at $n=4$: 8 of the 64 signatures orientable ($p_4 = 1/8 = 2^{-3}$), $\mathbb{E}[F \mid \text{non-or.}] = 13/8$ and $\mathbb{E}[F \mid \text{or.}] = 9/4$, consistent with $\mathbb{E}F_4 = 109/64$.

B.3 Reference check

The writeup is **self-contained**: it invokes no named external theorem and cites no paper. The only inherited object is the flag/ribbon representation from the previous attempt reproduced in `prompt.md`, and the writeup re-proves it (S1–S4 above, all verified here independently).

- **Source conjecture**: confirmed verbatim against the arXiv v3 TeX source (<https://arxiv.org/e-print/2211.01032v3>, RE-arxiv3.tex 1.2604 and the model definition at 1.2601), quoted in full under *Interpretation*. The neighbouring conjecture ($\mathbb{E}[F^-] \leq \mathbb{E}[F]$ for all G) is consistent with the measurements here: non-orientable $\mathbb{E}F(K_{50}) = 4.15$ against the orientable ≈ 6.98 reported in the sibling verification.
- **Priority check (step 2b)**: the literature on random 2-cell embeddings is tiny and I enumerated it. arXiv full-text API queries for "random 2-cell embedding" return exactly three papers (2103.05036, 2202.07746, 2211.01032); "average genus" returns 17, the newest graph-theoretic one being 2211.01032 itself; "expected number of faces" returns 9, newest relevant 2211.01032 (2024-11's 2411.19163 is convex geometry). Four web searches (Sept 2026) for follow-ups, citing work, and 2026 arXiv listings found nothing addressing the non-orientable conjecture. The load-bearing general lemma (the $\mathcal{H}_d$ exposure lemma) is a folklore deferred-decision statement already used in the source paper's orientable analysis, not an independently publishable named result. **Not ALREADY KNOWN**.
- **Self-contamination check (step 2c)**: none of the searches returned this repository, its pull request, or any paraphrase of this campaign's writeups; the only hits were the three genuine arXiv papers above plus the source's SODA 2024 proceedings version. No claim of prior art is recorded here, so there is nothing that could be traced back to our own artifacts.

B.4 Computational check

Scripts in `verification_astra/scripts/2211.01032__03/` (Python 3.13, numpy 1.26; `model.py`, `exposure.py`, `exact_small.py`, `orient_check.py`, `hmodel_equiv.py`, `hit_avoid_small.py`, `trace_test.py`, `eq11_big.py`, `analyse_eq11.py`, `scaling.py`, `arith.py`). Everything below is from the **actual** model (uniform rotation + i.i.d. fair signature); the exposure lemma is never assumed when generating samples.

B.4.1 The flagged item: is equation (11) violated?

The apparent violation was reproduced at the earlier sample size — with 2000 replicates at $n=50$ the worst per- k ratio is 24.7, attained by a **single** observed good avoiding face at $k=278$ (bound 2.03×10^{-5} , empirical 5×10^{-4} , i.e. $1/2000$). Scaling up by 200x settles it.

Testing form (equivalent to (11) by symmetry of S under $\text{Aut}(K_n)_o$, since a good avoiding face of length k carries $2k$ flags of S): $\mathbb{E}G_k \leq C(n)q^{k-2}/k$ with $C(n) = (n-1)(n-2)/(2D^2)$.

n	reps	R	D	$C(n)$	good avoiding faces observed	worst bin ratio	worst single- k ratio	violated bins
50	400 000	23	25	1.8816	488 277	0.2698	2.79 (count 1, none $k=408$)	
100	40 000	27	71	0.9623	62 144	0.5704	16.07 (count 1, none $k=668$)	
200	12 000	31	167	0.7064	22 760	0.7251	145.8 (count 1, none $k=1349$)	

Binned (Poisson 99.9% two-sided limits on the counts; a bin is violated iff the **lower** limit exceeds the bound). Every bin, over the entire observed length range $3 \leq k \leq 1349$:

- $n=50$: ratios 0.2698, 0.2545, 0.2539, 0.2558, 0.2550, 0.2545, 0.2523, 0.2522, 0.2372, 0.2079, 0.2100, 0.1218, 0.1185, 0, 0, 0 over bins $[3, 5)$, $[5, 10)$, $\dots$, $[1000, \cdot)$. Predicted slack $(D/(n-2))^2 = (25/48)^2 = 0.2713$.
- $n=100$: 0.5263, 0.5051, 0.5035, 0.5067, 0.5083, 0.5016, 0.5239, 0.5036, 0.4960, 0.5051, 0.5093, 0.4791, 0.5704, 0.3114, 0, 0. Predicted slack $(71/98)^2 = 0.5249$.
- $n=200$: 0.7168, 0.6845, 0.6995, 0.7132, 0.7096, 0.7042, 0.6634, 0.7251, 0.7155, 0.6591, 0.6349, 0.6790, 0.6536, 0.6122, 0.6733, 0.6553. Predicted slack $(167/198)^2 = 0.7114$.

So the observed value is, uniformly in k and at three values of n , equal to the bound times exactly the slack $(D/(n-2))^2$ that the two last-step factors $1/D$, $1/(2D)$ (versus their true values $\approx 1/(n-2)$, $1/(2(n-2))$) predict — agreeing to three significant figures at every n , and drifting *down* at the largest k as the extra survival slack in (5) kicks in. The bound is asymptotically sharp but never violated. The specific event that raised the alarm is now measured: the bin $[250, 300)$ at $n=50$ has 99 observations and ratio 0.2100, and $k=277$ alone has 5 observations and ratio 0.602. **The apparent violation was Monte-Carlo noise** — a single Poisson event whose bound-times-reps expectation was 0.34.

The same phenomenon reappears at every n as an artefact of the far tail (at $n=200$, one face at $k=1349$ gives a nominal ratio of 146 against a bound of 0.0069 expected counts); its 99.9% Poisson lower limit is 0.001, i.e. no evidence whatsoever.

B.4.2 Other inequalities of the writeup, same runs

n	$\mathbb{E}F^{\text{hit}}$	bound (2)	$\mathbb{E}F^{\text{avoid}}$	bound (12)	$\mathbb{E}B$	bound (10)
50	2.9276	3.2294	1.2207	4.7549	0.000000	0.02
100	3.2832	3.5836	1.5536	3.0602	0.000000	0.01
200	3.6318	3.9340	1.8967	2.7155	0.000000	0.005

A notable confirmation of *sharpness*, not of error: the writeup’s asymptotic form $\mathbb{E}F^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4}$ evaluates to 1.2062 at $n = 50$ and 1.5526 at $n = 100$, against measured 1.2207 and 1.5536 — the chain of inequalities for the avoiding faces is an asymptotic *equality* (the measured values exceed the limit form only by the positive $O(\ln n/n)$ correction that the theorem’s error term allows, and stay far below the proved finite- n bound). $\mathbb{E}F^{\text{hit}}$ sits a constant ≈ 0.30 below (2), the slack coming from the “+1” for the forced last pair.

B.4.3 Global sanity check of the theorem itself

$\mathbb{E}F_n$ in the non-orientable model versus the claim $\leq \ln n + \frac{1}{4} + \frac{\gamma}{2} = \ln n + 0.5386$:

n	reps	$\mathbb{E}F_n (\pm \text{s.e.})$	$\ln n + \frac{1}{4} + \frac{\gamma}{2}$	proved finite- n bound	$\mathbb{E}F_n - \ln n$
20	20 000	3.2403 $\pm$ 0.0101	3.5343	n/a ($D \leq 0$)	0.2446
50	20 000	4.1615 $\pm$ 0.0121	4.4506	7.9843	0.2495
100	8 000	4.8656 $\pm$ 0.0210	5.1438	6.6439	0.2605
200	3 000	5.5217 $\pm$ 0.0375	5.8369	6.6495	0.2233
400	800	6.3975 $\pm$ 0.0793	6.5301	7.0051	0.4060
800	200	6.9150 $\pm$ 0.1673	7.2232	7.5076	0.2304
1600	60	7.6167 $\pm$ 0.3275	7.9164	8.0871	0.2389

$\mathbb{E}F_n - \ln n$ is flat at ≈ 0.24 over two decades of n — the conjecture is true empirically with a constant comfortably below the writeup’s 0.5386, and the 0.30 measured slack in (2) accounts almost exactly for the difference ($0.5386 - 0.30 = 0.24$). Independent 400 000-replicate run at $n = 50$ gives $\mathbb{E}F_{50} = 4.1483$, consistent.

B.5 Caveats

- (11)’s **conditioning is asserted, not proved**. “To estimate a good-face event, discard continuations whose histories cease to satisfy the load restriction” is the whole argument. The repair is routine (given in S14 above) but has one step that is not purely mechanical: applying the $1/(2D)$ bound at the last query needs $r_{w_0} \leq R$ after query $k-1$, which holds only because the trace’s active vertex changes at every step, so query $k-1$ loads $u \neq w_0$. A referee should require this paragraph to be written out.
- No explicit n_0** . “For sufficiently large n ” in (7) hides $D = n - 2 - \lfloor 4 \log_2 n \rfloor > 0$, which fails up to $n \approx 32$ ($n = 20$ gives $D = 0$, $n = 32$ gives $D = 10$ and $C(n) = 68$). Harmless — the conjecture is asymptotic and $F \leq n(n-1)$ always — but the writeup should name the threshold.
- (10) is exactly at the boundary**. $n^{3/2-R} \leq 1/n$ holds only because R is a *ceiling* of $4 \log_2 n$; with $R = \lfloor 4 \log_2 n \rfloor$ the chain would fail.
- Simplicity of K_n is load-bearing and unstated as a hypothesis**. (4) and the “at most two flags to w_0 ” step in (11) use that there is one edge between any two vertices; the argument does not transfer verbatim to multigraphs (where the source’s companion conjecture $\mathbb{E}[F^-] \leq \mathbb{E}[F]$ also lives).
- The model-reduction sentence** “with an immaterial fixed adjustment if the opposite sign convention is used” is hand-waved. It is correct (verified here to TV distance ≤ 0.0026 at $n = 4, 5, 6$), but a reader cannot check it as written.
- In §2 the exposure bound is applied to a flag identified by a different path structure** ($P \cup Q_o$) than the one the lemma forbids ($J_o \cup Q_o$). The word “at most” saves it, but the writeup does not point this out.
- The boxed theorem is asymptotic**. At $n = 50$ the argument actually proves only $\mathbb{E}F_{50} \leq 7.98$ (against the display’s 4.45); the constants $\frac{1}{4} + \frac{\gamma}{2}$ are limits. This is consistent with the claim as stated (“As $n \rightarrow \infty$ ”) but the verdict block’s one-line phrasing could be read as a finite- n statement.
- Edge cases**. $n = 1, 2$: $\mathbb{E}F_2 = 1$, trivial; $k = 0, 1, 2$ face lengths handled (S15); disconnected graphs do not arise. $d = 1$ ($n = 2$) breaks $\mathcal{H}_d$ ’s no-proper-cycle argument, which is why S15 assumes $n \geq 3$ — correctly stated.
- Not independently refereed by a human**. The claim resolves a conjecture of Campion Loth–Halasz–Masařík–Mohar–Šámal that is about 1.5 years old; the argument is short and, by the measurements above, asymptotically tight in its avoiding-face half. That combination is exactly what one would want an expert pass on before treating it as settled.

B.6 Referee summary

The one item flagged as potentially decisive — an apparent violation of equation (11) at $n = 50$ — is **Monte-Carlo noise**, and I can say so with numbers. With 400 000 replicates at $n = 50$ (488 277 good avoiding faces), 40 000 at $n = 100$ and 12 000 at $n = 200$, the empirical $\mathbb{E}G_k$ sits at 0.27, 0.50 and 0.70 of the bound respectively, uniformly across every length bin from $k = 3$ to $k = 1349$ — matching to three significant figures the slack $(D/(n-2))^2 = 0.2713, 0.5249, 0.7114$ that the inequality’s own two last-step factors predict. Not one bin is violated; the “worst” per- k ratios (2.8, 16, 146) are all single-count Poisson events in the far tail whose 99.9% lower confidence limits are $\approx 10^{-3}$, exactly the artefact that the 2000-replicate pilot hit at $k = 278$. Independently of the simulation, (11) is a rigorous consequence of the exposure lemma: I supplied the nested-conditioning argument the writeup only gestures at, and it goes through.

Beyond that, I re-derived every step. The model reduction to i.i.d. uniform $\mathcal{H}_d$ local matchings is correct (and the per-incidence bit trick is the right one — a per-vertex switching would have given only coboundaries; $|\mathcal{H}_d| = 2^{d-1}(d-1)!$) and the exposure lemma were verified by exhaustive enumeration; the fixed-vertex split, the escape bound (3)–(5), the congestion supermartingale (6), the bad-face

bound (9)–(10), the summation (12)–(15) and the final constants all check out symbolically and numerically, and $\mathbb{E}F_n - \ln n$ measured over $n = 20 \dots 1600$ is flat at ≈ 0.24 , below the claimed $\frac{1}{4} + \frac{\gamma}{2} = 0.539$ by almost exactly the 0.30 slack I measure in inequality (2). The result is not in the literature (the whole subject is three arXiv papers). I found **no mathematical error**. I record MINOR_GAPS rather than CONFIRMED for the expositional deficits listed above — chiefly that the proof of (11), the single most load-bearing inequality, is compressed into one sentence, and that no large- n threshold is named — and because a claimed resolution of a live conjecture, reached in one unrefereed pass, deserves a human expert’s eyes on §4.2 before it is treated as settled.

References

- [1] J. Campion Loth, K. Halasz, T. Masařík, B. Mohar and R. Šámal, Random embeddings of graphs: the expected number of faces in most graphs is logarithmic, [arXiv:2211.01032](#) (v3, April 2025); extended abstract in *Proc. SODA 2024*, 1177–1195. Conjecture 9.5 and the definition of the non-orientable random embedding are in Section 9.
- [2] Graph Theory AI project, The expected number of faces of a random embedding of a dense simple graph is $\Theta(\log n)$, machine-generated note on Conjecture 9.3 of [1], catalog id 2211.01032__02, to_review/2211.01032__02__random-embeddings-expected-faces-Theta-log-n_note.pdf in [this repository](#) (September 2026); referee report in verification/2211.01032__02.md.
- [3] B. Mohar and C. Thomassen, *Graphs on Surfaces*, Johns Hopkins University Press, 2001, Chapter 3 (embedding schemes, face tracing, orientability).
- [4] J. L. Gross and T. W. Tucker, *Topological Graph Theory*, Wiley, 1987 (reprinted Dover, 2001), Chapter 3 (rotation systems and their non-orientable generalisation).